

Opioids

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Session Objectives

- 1. Describe the opioid analgesic drug classes; including full and partial agonists.**
- 2. Identify the specific drugs. Correlate the use of each drug to the mechanism of action.**
- 3. Describe the specific adverse effects associated with the use of opioids.**
- 4. Identify how differences in pharmacokinetics can enhance adverse effects.**
- 5. Recognize the signs and symptoms of an opioid overdose. Differentiate an opioid overdose from withdrawal.**
- 6. Describe the critical use of opioid antagonists in overdose situations.**

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Opium Extraction

Opium from the Poppy (Papaver Somniterum) Plant found in Turkey, Mexico, Southeast Asia, China, and India . Blooms in June, ten days later, incisions are made in the capsules and a milky fluid oozes out. Gummy mass (now brown) is scraped off and pressed into cakes of raw opium to dry.

Opium contains over 20 compounds: morphine (10%) and codeine (0.5%). Morphine is extracted in a relatively pure form. Codeine is low concentration, and is synthesized from morphine by an ether-type methylation.

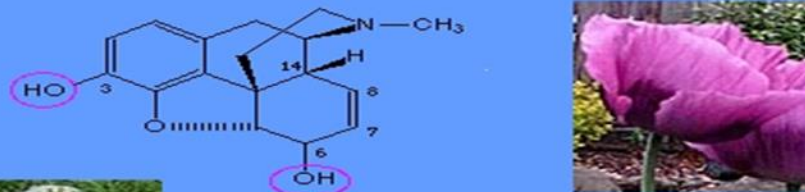


Primary Natural Opioids

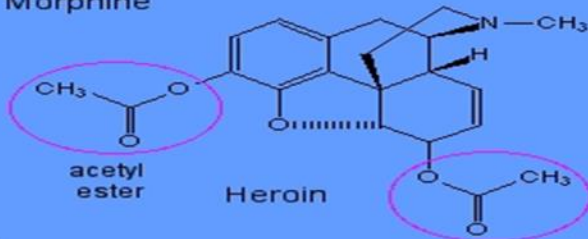
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Narcotic Analgesics



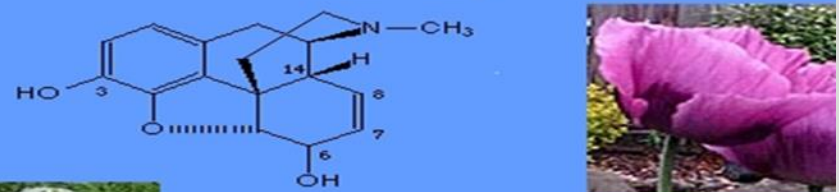
Morphine



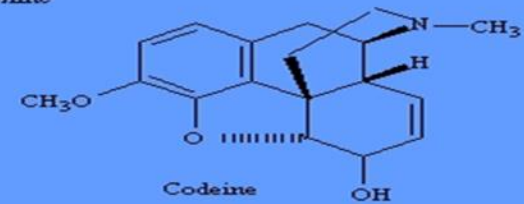
Heroin

C. Ophardt, c. 2003

Narcotic Analgesics



Morphine



Codeine

C. Ophardt, c. 2003

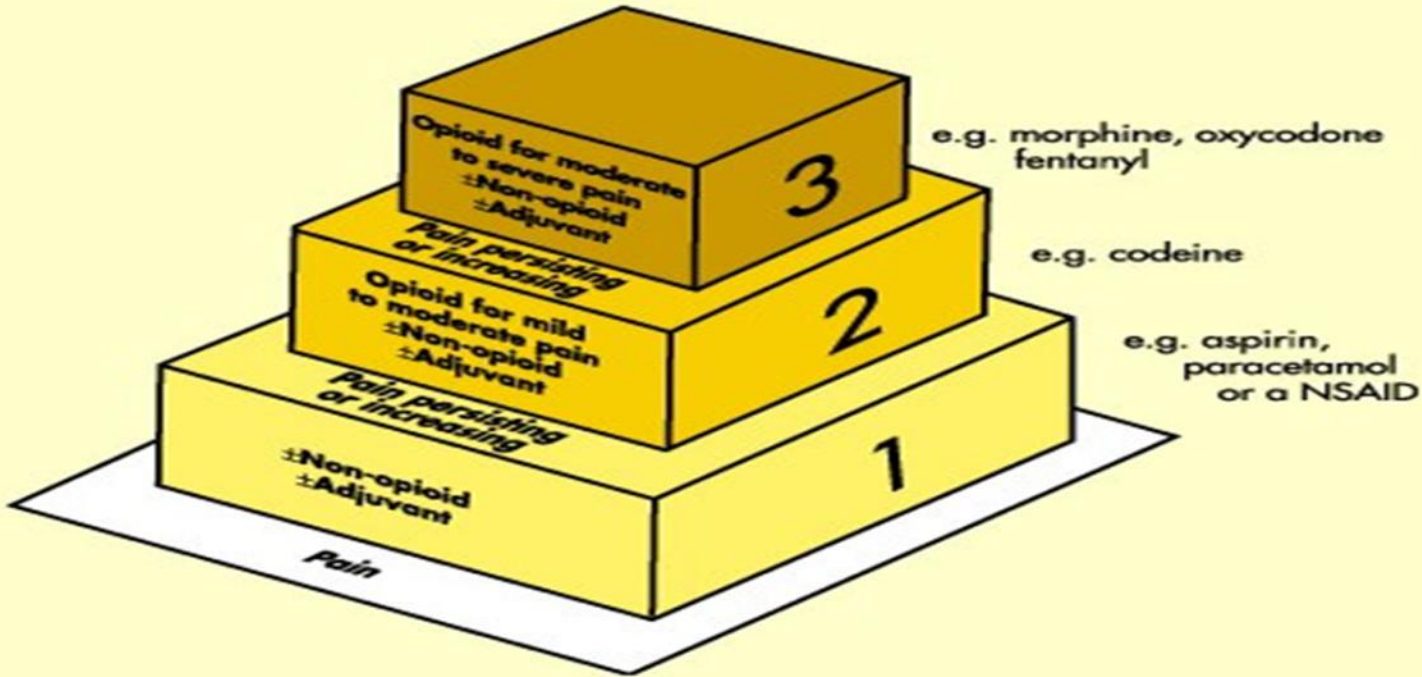
Papaverine-like

Source: C.E. Ophardt, Elmhurst College, Virtual Chembook

World Health Organization

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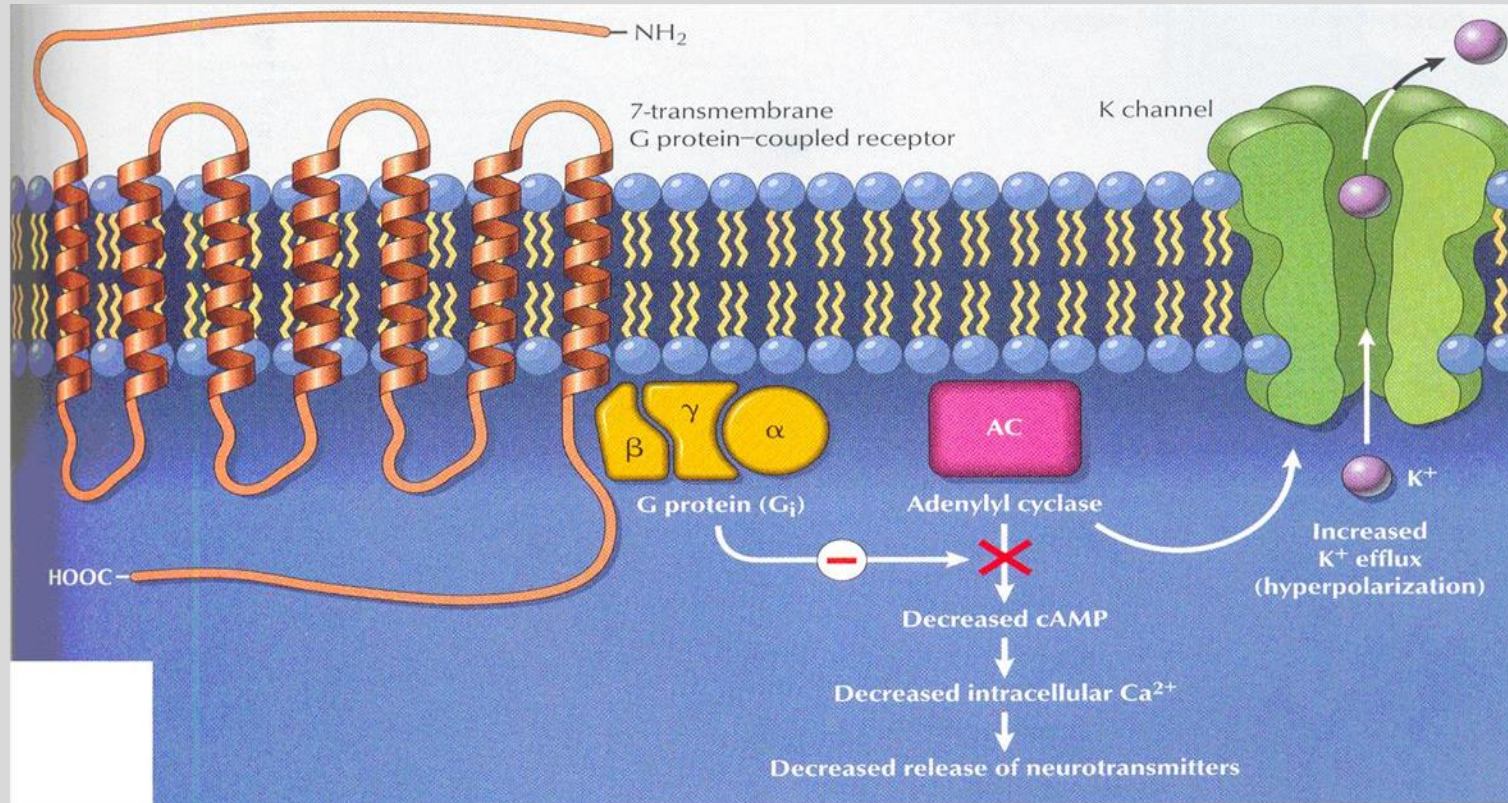


Opioid Receptors

- δ : analgesia, anxiety/mood
- μ : analgesia, euphoria, physical dependence, bradycardia, GI, respiratory depression, and miosis
- κ : analgesia, sedation, hallucinations, dissociation, and dysphoria.

Opioids: Gi Receptor Mechanism

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Pharmacokinetics

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- Well absorbed, rapid distribution.
- Highly lipophilic
- First pass effect
- Glucuronidation to metabolites: morphine-6-glucuronide, normeperidine.
- Renal excretion
- Patient controlled analgesia (PCA) and spinal delivery

Analgesia

- **Strong agonists: morphine, methadone, heroin, meperidine, fentanyl**
- **Mixed agonist-antagonist: buprenorphine**
- **Weak (or partial) agonist: codeine, hydrocodone (often combined with COX inhibitor).**

Cardio-Respiratory

- **Vasodilation= hypotension**
- **Reduction of myocardial oxygen demand, cardiac work.**
- **Decrease PVR, preload, afterload**
- **Cause of death in overdose =Respiratory depression**
- **Opioids decrease the sensitivity of the medulla to increased carbon dioxide.**
- **Increase intracranial pressure (worsen head injury outcome).**

Gastrointestinal

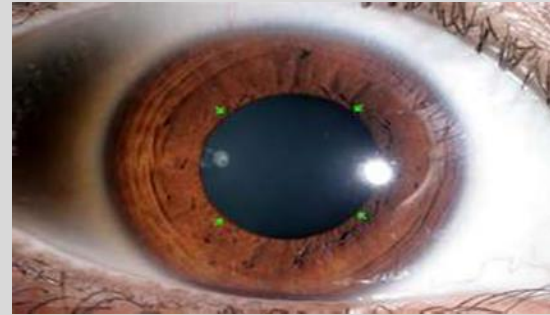
- Constipation (inhibit motility)
- **Encourage docusate, fiber, fluid**
- Constriction of sphincter of Oddi, urinary sphincter
- Uterus: Prolong labor
- Emesis (chemotrigger zone, CTZ)

Eye

- Parasympathetic effect
- Stimulation of the Edinger-Westphal nucleus of oculomotor nerve III
- Miosis is characteristic of the use and abuse of the opioids.
- No tolerance is observed (diagnostic).



Miosis



Mydriasis

Other effects

- **Histamine release (pruritis, bronchospasm, vasodilation)**
- **Increased prolactin release**
- **Inhibition of LH, FSH release**
- **Suppress immunity**
- **Adjunctive in anesthesia (Alfentanil, Sulfentanil, Remifentanil).
Provides sedation and analgesia**

Neonatal Abstinence Syndrome

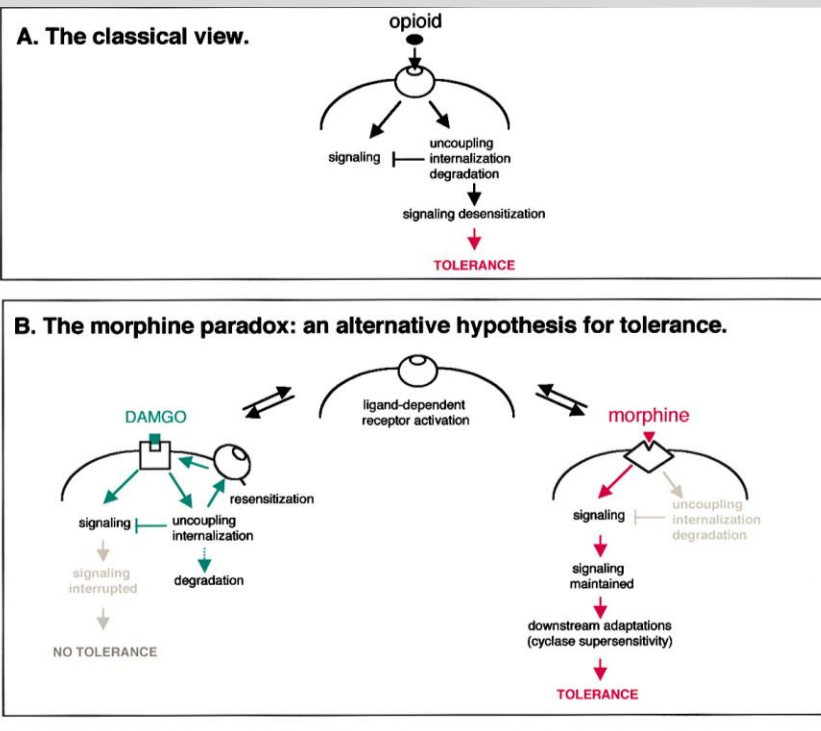
- The fetus may become physically dependent in utero and manifest withdrawal symptoms early postpartum
- Withdrawal symptoms in the infant: irritability, shrill crying, diarrhea, or seizures.
- More severe withdrawal: camphorated tincture of opium (previous). Current: Morphine, methadone
- Phenobarbital and clonidine (adjunctive use)

Tolerance: Changing Theories

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- Occurs with all opioids
 - Receptor uncoupling and degradation (fewer receptors on surface)
 - Dysfunction of structural interactions between the μ receptor G proteins, second-messenger systems
- **No tolerance to miosis or constipation**
- This (and dependence) should not be used as a barrier to providing patients with the best possible care and quality of life.



Opioid Withdrawal

- Occurs 6-12 h after last dose, continues for several days, and ends within 72 h.
- Symptoms: Rhinorrhea, sweating, lacrimation, chills, gooseflesh, yawning, anxiety, aggression, mydriasis
- Drugs with a long $t_{1/2}$, produce a prolonged withdrawal (Methadone)
- Opioids bind in the nucleus accumbens, release dopamine leading to addiction “reward center”

Hyperalgesia

- **Increase the sensation of pain**
- **Activation of the bradykinin and NMDA receptors are likely causes.**
- **Discourage opioids for chronic pain**

Morphine

- “Standard” opioid agonist
- Reacts with all opioid receptors
- Uses: moderate-severe pain, reduces dose of anesthetics, and anxiolytic effect.
- MI: produces vasodilation
- Hypersensitivity in respiratory disease
- **Respiratory depression (fatal)**
- Not for head trauma patients
- Intense sedation and respiratory depression when combined with other **sedatives, alcohol, or barbiturates.**

Codeine

- Mild-moderate pain; metabolized to morphine. Considered to have “partial” activity.
- Combined with non-opioids to relieve pain.
- Antitussive
- Codeine undergoes **CYP2D6** metabolism to morphine. Genetic polymorphisms may increase effects.
- “Ultra-fast metabolizer” is a concern.

Meperidine

- **Rapid onset-short duration of action.**
- **Normeperidine (metabolite) can induce CNS stimulation.**
- **Serotonin syndrome with SSRIs, linezolid, tricyclic antidepressants, “triptan” drugs for migraines.**
- **Not with monoamine oxidase inhibitors (MAOIs)**
- **Anti-muscarinic effects (mydriasis, tachycardia)**
- **Some local anesthetic effects**

Serotonin Syndrome

- SSRIs, TCAs, meperidine, dextromethorphan, St. John's Wort, linezolid, "triptans"
- Symptoms: rigidity, hyperthermia, myoclonus, delirium, hypertension, tachycardia within 24 hrs, unlike neuroleptic malignant syndrome (weeks to months). Also, GI effects
- Treatment: Benzodiazepines, cyproheptadine, ventilation

Diphenoxylate

- **Antidiarrheal**
- **Atropine reduces abuse potential**
- **Loperamide: derivative of diphenoxylate. Does not cross BBB.**

Fentanyl

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- **80-100x more potent than morphine; transdermal or transmucosal. Lipophilic**
- **Every 3 days (patch)**
- **Lozenges to induce anesthesia in children.**
- **Contraindicated in pregnancy**

Levorphanol

- **L-isomer of morphine**
- **Moderate to severe pain**
- **D-isomer (Dextromethorphan; DM). Not addictive
cough suppression at normal doses.**
 - **NMDA receptor antagonist**
 - **SERT and NET reuptake inhibition**

Methadone

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- Like morphine, but has better bioavailability and **long duration ($t_{1/2}$ = 15-40h)**
- Treats heroin addiction.
- CYP 3A4 substrate
- NMDA receptor antagonist
- Moderate to severe pain

Heroin: Schedule I

- **Injected, snorted, adulterated.**
- **Chronic injection leads to collapse of vessels**
- **Highly lipophilic**
- **Metabolized to morphine ~diacetylmorphine**
- **Pregnancy: newborn addicted to heroin, withdrawal.**
- **Increased incidence of HIV and hepatitis (needle sharing)**
- **Oxycontin[®]- gave a “heroin like” effect when crushed and snorted. New formulations provide special coatings to prevent crushing.**

Tramadol

- μ receptor agonist and reuptake inhibitor of NE and 5HT
- Can precipitate withdrawal.
- CYP2D6 polymorphism “ultra-fast metabolism”
- Used for chronic pain (neuropathic)
- Combined with acetaminophen/NSAIDs
- Tapentadol (μ receptor agonist and reuptake inhibitor of NE, minimal effect on 5-HT).
- Both have serotonin syndrome warning.

Methylnaltrexone

- Treats opioid-induced constipation.
- Does not cross blood brain barrier (BBB). Quaternary ammonium cation
- Alvimopan: treats postoperative ileus. Antagonist of mu receptor in GI tract. Contraindicated in patients who had opioids within 7 days; do not give alvimopan for more than 7 days. MI risk increased.

Mixed Agonist-Antagonist

- Pentazocine, Butorphanol, Nalbuphine
- Retains analgesia with reduced addiction potential ($\kappa > \mu$ receptor).
- Less physical dependence and respiratory depression.
- Psychomimetic effects (κ receptor). Pentazocine
- **Precipitate withdrawal in morphine addicted patient.**

Buprenorphine

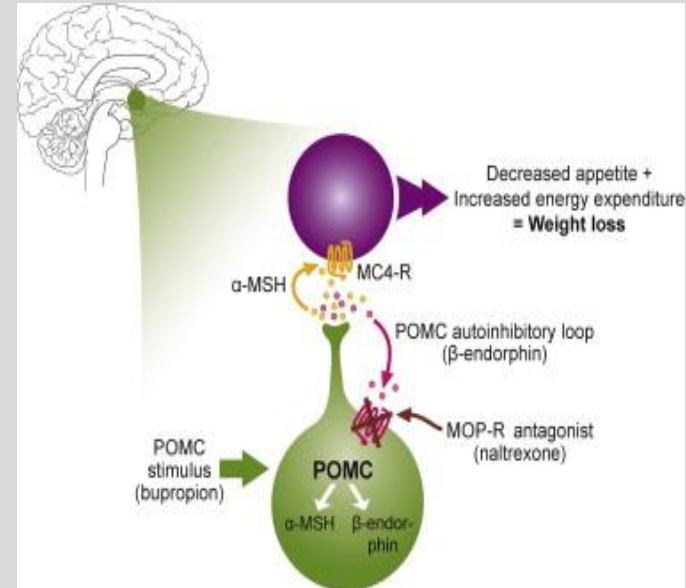
- Partial μ agonist. At higher doses is an antagonist because of K receptor blockade
- Slow dissociation from μ receptor.
- CYP 3A4 substrate
- Sublingual, transdermal, parenteral
- Adverse effects are similar to morphine and pentazocine
- Treats opioid dependence
- Precipitate withdrawal in opioid users
- Buprenorphine and naloxone (Suboxone[®])

Opioid Antagonists

- **Naloxone: IV; fast onset for opioid overdose (1-3 min). Rapid offset. Repeated administration until opioid is cleared.**
- **Approved in neonates with respiratory depression born to heroin mothers**
- **Naltrexone: oral; long-duration. Maintenance therapy. Reduces alcohol cravings. Combined with bupropion for weight loss potential.**

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Opioid Abuse

- Heroin: Rapid tolerance and dependence
- MOA: μ and κ receptor agonism
- Overdose: IV naloxone and ventilation
- Symptoms: “everything is down”, stupor, bradycardia, decreased respirations, miosis.
- Withdrawal: Methadone or buprenorphine
- Symptoms: lacrimation, rhinorrhea, yawning, sweating, gooseflesh, tremors.

DEA DRUG SCHEDULING

Schedule I

Substances defined as drugs with no currently accepted medical use and a high potential for abuse.



Schedule II

Substances defined as drugs with a high potential for abuse, with use potentially leading to severe psychological or physical dependence.



Schedule III

Substances defined as drugs with a moderate to low potential for physical and psychological dependence.



Schedule IV

Substances or chemicals defined as drugs with a low potential for abuse and low risk of dependence.



Schedule V

Substances defined as drugs with lower potential for abuse. Schedule V drugs are generally used for antidiarrheal, antitussive, and analgesic purposes.



Examples

Marijuana (Cannabis)
Ecstasy
Heroin
LSD
Methaqualone (Quaaludes)
Peyote

Hydrocodone (Vicodin)
Cocaine
Meth/Fentanyl
Methadone
Hydromorphone (Dilaudid)
Meperidine (Demerol)
Oxycodone (OxyContin)
Adderall/Ritalin

Codeine (less than 90 mg per dosage/Tylenol)
Ketamine
Anabolic steroids
Testosterone

Xanax
Soma
Darvocet
Valium
Ativan
Talwin
Ambien
Tramadol

Robitussin AC (cough meds)
Lomotil
Motofen
Lyrica
Parepectolin

Source: dea.gov

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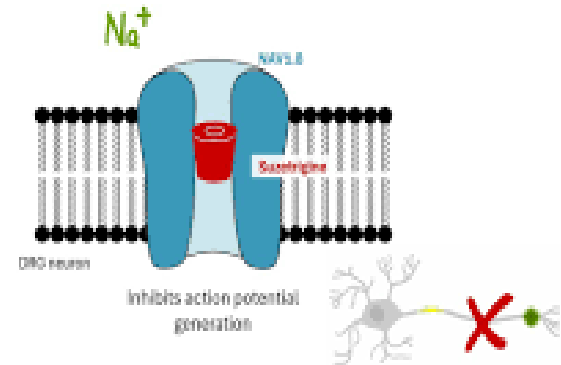
Suzetrigine

- **Non-opioid**
- **MOA: Sodium channel blocker, works in periphery**
- **Use: moderate-to-severe acute pain, not more than 14 days.**
- **Adverse effects: itch, muscle spasms (increased creatine kinase), N/V.**

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Suzetrigine is a potent and selective inhibitor of NaV1.8



Summary Slide

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- Describe opioid analgesic drug classes; including full and partial agonists
- Describe the critical use of opioid antagonists in overdose situations.

Core References:

- Basic & Clinical Pharmacology 15e, 2024. Bertram G. Katzung, Todd W. Vanderah; Chapter 31: Opioid Agonists & Antagonists
- Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 23: Opioid Analgesics
- Opioids Summary guide and Practice questions (Maria A. Pino, Ph.D)

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>